Skills Summary

Area Models for Two-Digit Multiplication

Use this reference while completing your worksheet or when studying.

Skill 1 — Building the area model

Rule: Split each factor into *tens* + *ones*, put one factor across the top and the other down the side, and multiply every row label by every column label. Four regions, four partial products, one sum.

	tens	ones
tens	T×T	O×T
ones	T×O	O×O

Example: Multiply 21×13.
Step 1. Split by place value — $20 + 1$ and $10 + 3$ — because the model needs one column and one row for each place.
Step 2. Regions: $20 \times 10 = 200$, $1 \times 10 = 10$, $20 \times 3 = 60$, $1 \times 3 = 3$. Adding all four: **273**

Try it: Multiply 34×12 with an area model. check: **408**

Skill 2 — Estimating the product first

Rule: Round *each* factor to the nearest ten, then multiply the rounded factors. The estimate tells you how big the rectangle should be, so a wildly different total means a region was missed.

Example: Estimate 27×43.
Step 1. Both factors are being rounded, so look at each ones digit on its own — that is what decides up or down.
Step 2. $27 \to 30$ and $43 \to 40$, so the estimate is $30 \times 40 =$ **1200**

Try it: Estimate 52×38 by rounding each factor to the nearest ten. check: **2000**

Skill 3 — Finding a missing partial product

Rule: Every region is (row label) × (column label) — so an empty region can always be rebuilt from its own two labels. Never total a model until all four regions are filled.

Example: One region of the model for 43×25 is blank: columns 40 and 3, rows 20 and 5, with 800, 200 and 60 already written.
Step 1. The blank sits where row 5 meets column 3, so its own labels rebuild it — no other region is needed.
Step 2. $3 \times 5 = 15$, and then $800 + 200 + 60 + 15 =$ **1075**

Try it: A model for 62×34 has columns 60 and 2, rows 30 and 4. Fill all four regions and total them. check: **2108**

(!) Watch out: Two regions, not four, is the most common error: 24×36 is *not* 20×30 plus 4×6. The two cross regions (tens meeting ones) are usually the biggest part of the answer.